Traffic Lights

Programming Steps: Step One: checking your connection:

1- Start by creating variables for each port used in the Arduino and name it with the road number and the light color. Example: GreenR1 (green Light for road one).

2- Define the pin modes in the void setup for all LED and push buttons.

3- Start the serial monitor.

4- In the void loop: Turn all the LEDs ON for 1 second and print the read values from both buttons on the serial monitor. Then turn OFF all LED and print the Values of the push buttons on the serial monitor.

Traffic Lights

<u>Challenge Details:</u>

1- The street is formed of three main roads having a T-shape.

2- Each road should have a red and green light.

3- The intersecting road should have two more lights green and red for pedestrians.

4- The intersecting road should be equipped with push buttons on both sides for pedestrians.

<u>Challenge rules:</u>

1- Cars can go between roads if both green lights are ON.

2- Cars can't go to the side road if the side lights are red.

3- The lights should cycle between the three roads as long as no pedestrians press the road button to pass.

4- Once a pedestrian presses the button, the red light for pedestrians should blink for 5 times then the green light should be ON while the red light of road 2 should turn ON.

Traffic Lights

Step Two: Programming the Road Cycle:

2- It requires 3 “if/else statements” nested in each other.

3- In order to control the time and be able to interfere with the process if a pedestrian presses the button, add 2 “for loops” nested in each other. Then add the “if/else statements” in them.

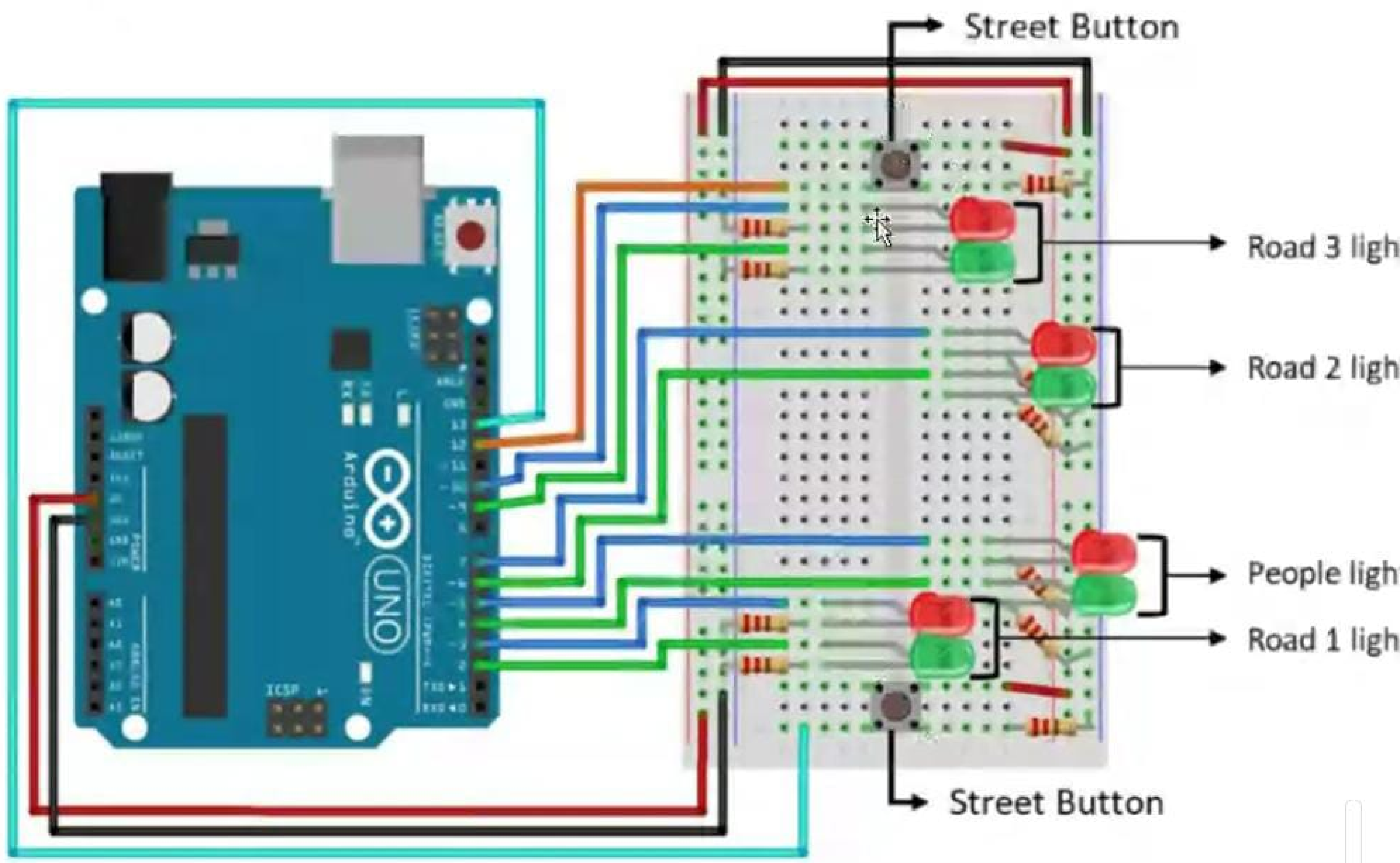
4- The first “for loop” will be counting between 0 and 3, such that its value will be used to decide which case will be done.

5- The Second “for loop” will be counting between 0 and 125 in order to keep the program cycling. Each cycle will have 40 ms delay. This will produce a total of (40 * 125 = 5000 ms) which is 5 seconds in each case.

6- The “if/else statements” will be added inside the second “for loop”.

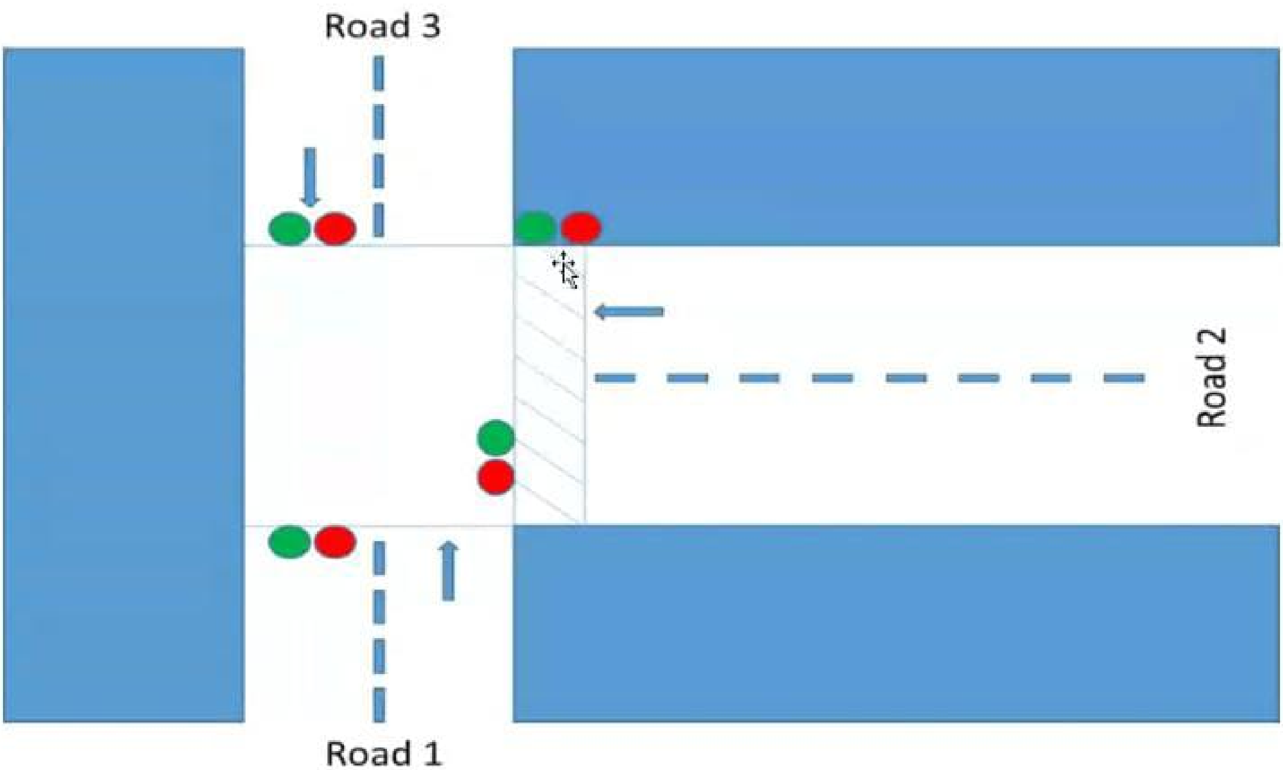
Programming

```
void loop() {
  // put your main code here, to run repeatedly:
digitalWrite(GreenR1, HIGH);
digitalWrite(RedR1, HIGH);
digitalWrite(GreenR2, HIGH);
digitalWrite(RedR2, HIGH);
digitalWrite(GreenR3, HIGH);
digitalWrite(RedR3, HIGH);
digitalWrite(GreenP, HIGH);
digitalWrite(RedP , HIGH);
Serial.print(digitalRead(12));
Serial.println(digitalRead(13));
delay(1000);
digitalWrite(GreenR1, LOW);
digitalWrite(RedR1, LOW);
digitalWrite(GreenR2, LOW);
digitalWrite(RedR2, LOW);
digitalWrite(GreenR3, LOW);
digitalWrite(RedR3, LOW);
digitalWrite(GreenP, LOW);
digitalWrite(RedP , LOW);
Serial.print(digitalRead(12));
Serial.println(digitalRead(13));
delay(1000);
}
```



Street Button
Road 3 ligh
Road 2 ligh
People ligh
Road 1 ligh
Street Button

Traffic Lights



dot.

Programming

```
} else if(i==2){ /*allowing cars between roads 1 and 2*/

  pedastrians();

digitalWrite(RedR1, LOW);
digitalWrite(RedR2, LOW);
digitalWrite(RedR3, HIGH);
digitalWrite(GreenR1, HIGH);
digitalWrite(GreenR2, HIGH);
digitalWrite(GreenR3, LOW);
digitalWrite(RedP, HIGH);
digitalWrite(GreenP,LOW);

delay(50);
```

sketch_jul28a.ino

```
const int Button1=13;

void setup() {
  // put your setup code here, to run
pinMode(RedR1,OUTPUT);
pinMode(RedP,OUTPUT);
pinMode(GreenR1,OUTPUT);
pinMode(GreenP,OUTPUT);

pinMode(Button1,INPUT);
Serial.begin(9600);
}

void loop() {
  // put your main code here, to run
digitalWrite(GreenR1,HIGH);
digitalWrite(GreenP,HIGH);
R1,HIGH);
digitalWrite(RedP,HIGH);
```

Mohamad is presenting

sketch_jul28a.ino

```
pinMode(Button2,INPUT);
Serial.begin(9600);
}

void loop() {
  // put your main code here, to run repeatedly:
digitalWrite(GreenR1,HIGH);
digitalWrite(GreenR2,HIGH);
digitalWrite(GreenR3,HIGH);
digitalWrite(GreenP,HIGH);

digitalWrite(RedR1,HIGH);
digitalWrite(RedR2,HIGH);
digitalWrite(RedR3,HIGH);
digitalWrite(RedP,HIGH);

Serial.println((digitalRead(Button1)==HIGH)? "Button 1 PRESSED": "Button 1 NOT PRESSED");
delay(1000);
Serial.println((digitalRead(Button2)==HIGH)? "Button 2 PRESSED": "Button 2 NOT PRESSED");
delay(1000);
}
```

TrafficLightsChallenge_Part2 | Arduino IDE 2.3.10

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Arduino UNO

TrafficLightsChallenge_Part2.ino

```
if(digitalRead(Button1)==HIGH || digitalRead(Button2)==HIGH){ // || : OR
for(int k=0;k<10;k++){
  digitalWrite(RedP,HIGH);
  digitalWrite(GreenR2,HIGH);
  delay(100);
  digitalWrite(RedP,LOW);
  digitalWrite(GreenR2,LOW);
  delay(100);
}
//Turn On the Green P and the RedR2 then turn of the GreenR2
digitalWrite(GreenP,HIGH);
digitalWrite(GreenR2,LOW);
digitalWrite(RedR2,HIGH);
digitalWrite(GreenR1,HIGH);
digitalWrite(GreenR3,HIGH);
digitalWrite(RedR1,LOW);
digitalWrite(RedR3,LOW);
delay(5000);
}
  }
}
```

Output

```
Sketch uses 1540 bytes (4%) of program storage space. Maximum is 32256 bytes.
```

Traffic Lights

Step Three: Interfering the process for pedestrians to pass:

1- In the Second "for loop" after the "if/else statements", we should add "if/else statement "that has the following condition: a. Condition: if button1 or button2 are pressed then start the interrupting process. (In order to include an OR condition in the "if statement" you should use the double straight slashes "||").

2- Add a "for loop" that repeats for five times in order to flash the red light for pedestrians (you can flash the green light for Road 2).

3- Out of the "flashing for loop" turn ON the pedestrians' green light, and stop the car in road

4- Delay for (2000 ms).

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Arduino UNO

TrafficLightsChallenge_Part2.ino

```
if(digitalRead(Button1)==HIGH || digitalRead(Button2)==HIGH){ // || : OR
for(int k=0;k<10;k++){
  digitalWrite(RedP,HIGH);
  digitalWrite(GreenR2,HIGH);
  delay(100);
  digitalWrite(RedP,LOW);
  digitalWrite(GreenR2,LOW);
  delay(100);
}
//Turn On the Green P and the RedR2 then turn of the GreenR2
digitalWrite(GreenP,HIGH);
digitalWrite(GreenR2,LOW);
digitalWrite(RedR2,HIGH);
/*OPTIONNAL digitalWrite(GreenR1,HIGH);
digitalWrite(GreenR3,HIGH);
digitalWrite(RedR1,LOW);
digitalWrite(RedR3,LOW);
delay(5000);*/
}
  }
}
```

Output

TrafficLightsChallenge_Part2.ino

```
}// ENDING OF THE ROAD CYCLES
delay(50);

if(digitalRead(Button1)==HIGH || digitalRead(Button2)==HIGH){ // || : OR
for(int k=0;k<10;k++){
  digitalWrite(RedP,HIGH);
  digitalWrite(GreenR2,HIGH);
  delay(100);
  digitalWrite(RedP,LOW);
  digitalWrite(GreenR2,LOW);
  delay(100);
}
//Turn On the Green P and the RedR2 then turn of the GreenR2
digitalWrite(GreenP,HIGH);
digitalWrite(GreenR2,LOW);
digitalWrite(RedR2,HIGH);
digitalWrite(GreenR1,HIGH);
digitalWrite(GreenR1,HIGH);
delay(5000);
}
  }
}
```


Building sketch   Ln 75, Col 1   Arduino UNO on COM6

```

void setup() {
  // put your setup code here, to run
pinMode(RedR1,OUTPUT);
pinMode(RedP,OUTPUT);
pinMode(GreenR1,OUTPUT);
pinMode(GreenP,OUTPUT);

pinMode(Button1,INPUT);
Serial.begin(9600);
}

void loop() {
  // put your main code here, to run r
digitalWrite(GreenR1,HIGH);
digitalWrite(GreenP,HIGH);
digitalWrite(RedR1,HIGH);
digitalWrite(RedP,HIGH);

Serial.println((digitalRead(Button1)==
delay(1000);
}
```

TrafficLightsChallenge_Part2 | Arduino IDE 2.3.10

File Edit Sketch Tools Help

Arduino UNO

TrafficLightsChallenge_Part2.ino

```
if(digitalRead(Button1)==HIGH || digitalRead(Button2)==HIGH){ // || : OR
for(int k=0;k<10;k++){
  digitalWrite(RedP,HIGH);
  digitalWrite(GreenR2,HIGH);
  delay(100);
  digitalWrite(RedP,LOW);
  digitalWrite(GreenR2,LOW);
  delay(100);
}
//Turn On the Green P and the RedR2 then turn of the GreenR2
digitalWrite(GreenP,HIGH);
digitalWrite(GreenR2,LOW);
digitalWrite(RedR2,HIGH);
/*digitalWrite(GreenR1,HIGH);
digitalWrite(GreenR3,HIGH);
digitalWrite(RedR1,LOW);
digitalWrite(RedR3,LOW);
delay(5000);*/
}
  }
}
```

Output

Traffic Lights

<u>Step Two: Programming the Road Cycle:</u>

1- In this step, use the "if/else Statement" to control the traffic Lights.

Case Number	Road 1 Lights	Road 2 Lights	Road 3 Lights	People Lights	Passes
Case 1	Red OFF Green ON	Red ON Green OFF	Red OFF Green ON	Red OFF Green ON	• Cars can go between road 1 and 3 • People can pass road 2
Case 2	Red OFF Green ON	Red OFF Green ON	Red ON Green OFF	Red OFF Green ON	• Cars can go between road 1 and 2 • People can't pass road 2
Case 3	Red ON Green OFF	Red OFF Green ON	Red OFF Green ON	Red OFF Green ON	• Cars can go between road 2 and 3 • People can't pass road 2

Traffic Lights

Programming Steps: Step One: checking your connection:

1- Start by creating variables for each port used in the Arduino and name it with the road number and the light color. Example: GreenR1 (green Light for road one).

2- Define the pin modes in the void setup for all LED and push buttons.

3- Start the serial monitor.

4- In the void loop: Turn all the LEDs ON for 1 second and print the read values from both buttons on the serial monitor. Then turn OFF all LED and print the Values of the push buttons on the serial monitor.

File Edit Sketch Tools Help

Arduino UNO

sketch_jul28b.ino

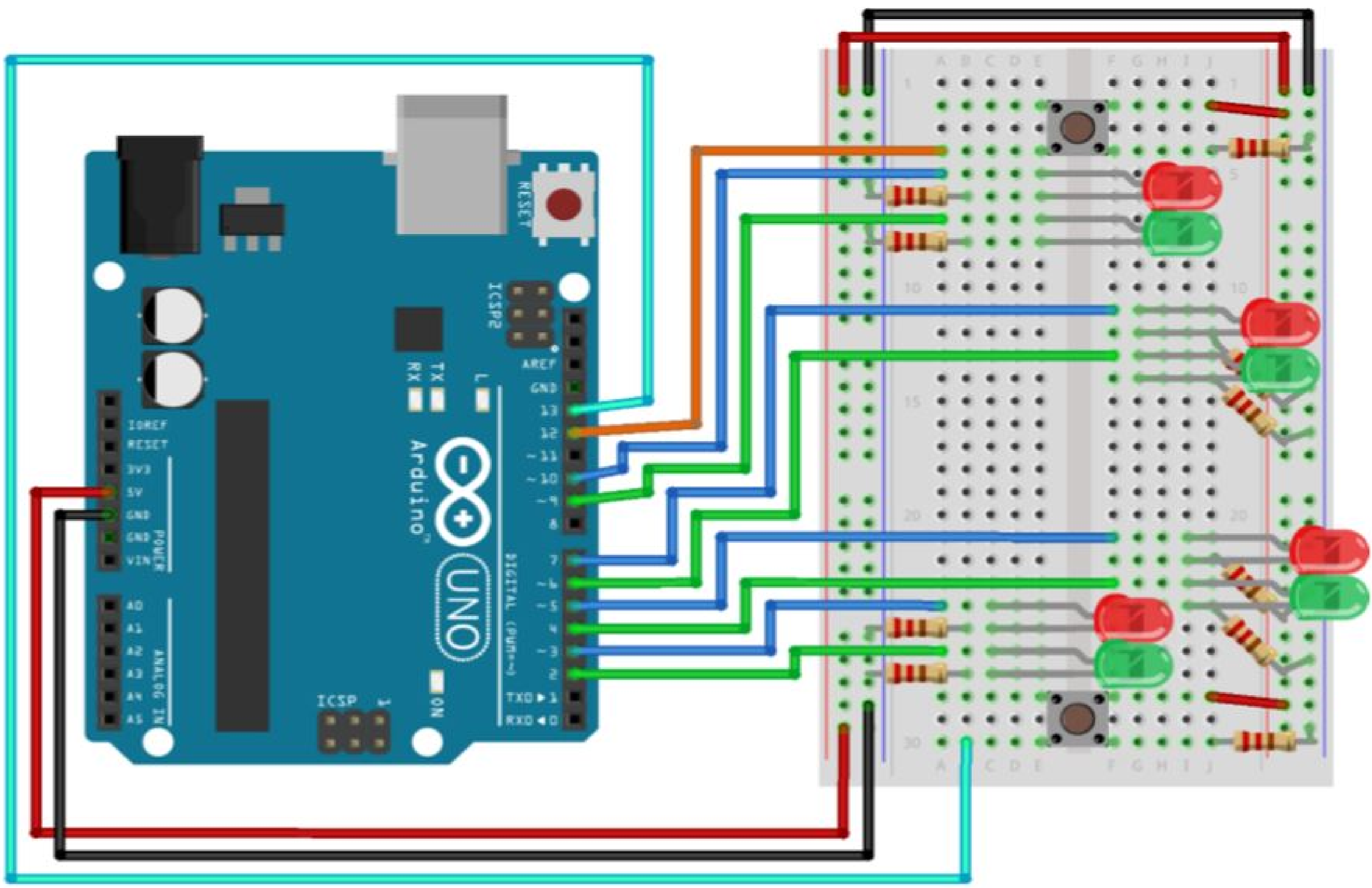
```
void setup() {
  // put your setup code here, to run once:
pinMode(RedR1,OUTPUT);
pinMode(RedR2,OUTPUT);
pinMode(RedR3,OUTPUT);
pinMode(RedP,OUTPUT);

pinMode(GreenR1,OUTPUT);
pinMode(GreenR2,OUTPUT);
pinMode(GreenR3,OUTPUT);
pinMode(GreenP,OUTPUT);

pinMode(Button1,INPUT);
pinMode(Button2,INPUT);
}

void loop() {
  // put your main code here, to run repeatedly:
for(int i=0;i<3;i++){ //LOOP TO SWITCH BETWEEN THE 3 CASES / THE 3 ROAD CYCLES
  for(int j=0;j<125;j++){ // LOOP TO GIVE A TIMING OF 40X125=5000 MS FOR EACH CY

  }
```



Arduino
UNO
RESET
ICSP2
ICSP
AREF
GND
IOREF
RESET
3V3
5V
GND
GND
VIN
POWER
A0
A1
A2
A3
A4
A5
ANALOG IN
DIGITAL (PWM=~)
TX
RX
L
ON
TXD
RXD

Traffic Lights

<u>Step Two: Programming the Road Cycle:</u>

2- It requires 3 “if/else statements” nested in each other.

3- In order to control the time and be able to interfere with the process if a pedestrian presses the button, add 2 “for loops” nested in each other. Then add the “if/else statements” in them.

4- The first “for loop” will be counting between 0 and 3, such that its value will be used to decide which case will be done.

5- The Second “for loop” will be counting between 0 and 125 in order to keep the program cycling. Each cycle will have 40 ms delay. This will produce a total of (40 * 125 = 5000 ms) which is 5 seconds in each case.

6- The “if/else statements” will be added inside the second “for loop”.

dot.

Case Number	Road 1 Lights	Road 2 Lights	Road 3 Lights	People Lights	Passes
Case 1	Red OFF Green ON	Red ON Green OFF	Red OFF Green ON	Red OFF Green ON	• Cars can go between road 1 and 3 • People can pass road 2
Case 2	Red OFF Green ON	Red OFF Green ON	Red ON Green OFF	Red O~~FF~~N Green O~~N~~FF	• Cars can go between road 1 and 2 • People can't pass road 2
Case 3	Red ON Green OFF	Red OFF Green ON	Red OFF Green ON	Red O~~FF~~N Green O~~N~~FF	• Cars can go between road 2 and 3 • People can't pass road 2

```
if(i==0){ //Case 1 allowing cars between roads 1
and 3
digitalWrite(RedR1,LOW);
digitalWrite(RedR2,HIGH);
digitalWrite(RedR3,LOW);
digitalWrite(RedP,LOW);

digitalWrite(GreenR1,HIGH);
digitalWrite(GreenR2,LOW);
digitalWrite(GreenR3,HIGH);
digitalWrite(GreenP,HIGH);
}
else if(i==1){ //Case 2 allowing cars between roads
1 and 2
digitalWrite(RedR1,LOW);
digitalWrite(RedR2,LOW);
digitalWrite(RedR3,HIGH);
digitalWrite(RedP,HIGH);

digitalWrite(GreenR1,HIGH);
digitalWrite(GreenR2,HIGH);
digitalWrite(GreenR3,LOW);
digitalWrite(GreenP,LOW);
}
else if(i==2){//Case 3 allowing cars between roads
2 and 3
digitalWrite(RedR1,HIGH);
digitalWrite(RedR2,LOW);
digitalWrite(RedR3,LOW);
digitalWrite(RedP,HIGH);

digitalWrite(GreenR1,LOW);
```

Case Number	Road 1 Lights	Road 2 Lights	Road 3 Lights	People Lights	Passes
Case 1	Red OFF Green ON	Red ON Green OFF	Red OFF Green ON	Red OFF Green ON	• Cars can go between road 1 and 3 • People can pass road 2
Case 2	Red OFF Green ON	Red OFF Green ON	Red ON Green OFF	Red O~~FF~~N Green O~~N~~FF	• Cars can go between road 1 and 2 • People can't pass road 2
Case 3	Red ON Green OFF	Red OFF Green ON	Red OFF Green ON	Red O~~FF~~N Green O~~N~~FF	• Cars can go between road 2 and 3 • People can't pass road 2

Challenge 2: Traffic Lights Simulation

```
const int RedR1=3,RedP=5, RedR2=7, RedR3=9;
const int GreenR1=2, GreenP=4, GreenR2=6,
GreenR3=8;
const int Button1=13, Button2=12;

void setup() {
  // put your setup code here, to run once:
pinMode(RedR1,OUTPUT);
pinMode(RedR2,OUTPUT);
pinMode(RedR3,OUTPUT);
pinMode(RedP,OUTPUT);

pinMode(GreenR1,OUTPUT);
pinMode(GreenR2,OUTPUT);
pinMode(GreenR3,OUTPUT);
pinMode(GreenP,OUTPUT);

pinMode(Button1,INPUT);
pinMode(Button2,INPUT);
}

void loop() {
  // put your main code here, to run repeatedly:
for(int i=0;i<3;i++){ //LOOP TO SWITCH BETWEEN
THE 3 CASES / THE 3 ROAD CYCLES
  for(int j=0;j<200;j++){ // LOOP TO GIVE A TIMING
OF 40X125=5000 MS FOR EACH CYCLE /
DELAY(40)
```

```
digitalWrite(RedP,HIGH);

digitalWrite(GreenR1,LOW);
digitalWrite(GreenR2,HIGH);
digitalWrite(GreenR3,HIGH);
digitalWrite(GreenP,LOW);
}// ENDING OF THE ROAD CYCLES
delay(50);

if(digitalRead(Button1)==HIGH ||
digitalRead(Button2)==HIGH){ // || : OR
for(int k=0;k<10;k++){
 digitalWrite(RedP,HIGH);
 digitalWrite(GreenR2,HIGH);
 delay(100);
 digitalWrite(RedP,LOW);
 digitalWrite(GreenR2,LOW);
 delay(100);
}
//Turn On the Green P and the RedR2 then turn of
the GreenR2
digitalWrite(GreenP,HIGH);
digitalWrite(GreenR2,LOW);
digitalWrite(RedR2,HIGH);
delay(5000);
}
 }
}
}
```